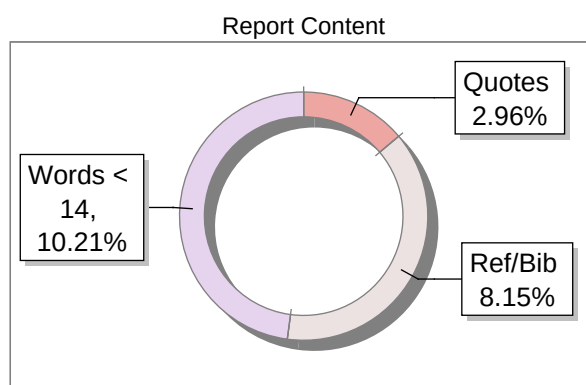
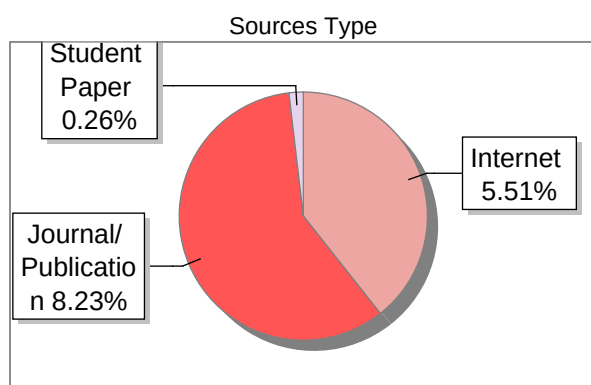


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Deep Learning-Based Tuberculosis Detection from Chest Radiographs

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Abstract—Tuberculosis (TB) continues to be one of the most dangerous respiratory diseases globally, causing significant morbidity and mortality annually. Quick and precise TB detection is crucial to prevent TB spread and manage patients appropriately. In this study, we propose a segmentation-free deep learning-based system to detect TB from chest X-ray images. Unlike previous methods, which perform pixel-wise segmentation to extract lung regions from images, our proposed technique utilizes transfer-learning based convolutional neural networks by feeding full chest X-ray images into the network, thus significantly reducing the complexity of the method without compromising the detection accuracy.

Since the number of labeled TB samples is limited, the class imbalance issue arises. To address this problem, Generative Adversarial Networks (GAN) are applied to generate new chest X-ray samples corresponding to the minority class of TB-positive images. Our proposed framework fine-tunes two popular backbone models, DenseNet121 and ResNet50, on the enlarged dataset. The models' accuracy, precision, recall, and F1 score are reported. Finally, an ensemble voting approach integrates the outputs of both models to enhance the stability of TB detection results. The experimental results show that the combination of DenseNet121 with generated TB samples achieves the best overall performance.

Index Terms—Tuberculosis detection, Chest X-ray imaging, Deep learning, Transfer learning, DenseNet121, ResNet50, Generative Adversarial Networks, Data augmentation, Medical image classification, Ensemble learning

I. INTRODUCTION

Mycobacterium tuberculosis, and continues to be one of the leading causes of death by infection in the world. The World Health Organization regards tuberculosis as a critical global health concern, with millions of cases occurring each year in spite of decades of efforts to control it. A timely bacteriological or radiological diagnosis of the condition is key in order for effective anti-tuberculosis treatment to follow, thus reducing transmission in the population and minimizing development of resistant strains.

Identify applicable funding agency here. If none, delete this.

Radiographic examination has proven essential in tuberculosis diagnosis in particular due to the ubiquity and inexpensiveness of the procedure relative to sputum culture testing or molecular analysis. However, radiograph reading is a very subjective process; accuracy highly depends on the level of expertise of the radiologist performing the evaluation, which makes it prone to error in areas with high prevalence and limited resources where patient throughput is high and specialist numbers are few. This problem has been instrumental in the drive towards automatic radiograph reading using AI.

In the realm of AI techniques, deep convolutional neural network (CNN) models have proven to be the leading approach for recognizing diseases from images. Pre-trained CNN models fine-tuned via transfer learning using domain-specific images typically perform better than classical machine-learning algorithms in a wide range of applications, including detecting lung nodules, diagnosing pneumonia, and identifying TB cases. While this may seem promising, one of the main problems encountered is that of the insufficient size and variety of publicly available labeled datasets of medical images, which often result in overfitting and suboptimal model generalization.

In our work, we address this issue by combining data synthesis using GANs with transfer learning. We train a GAN to generate images of TB-positive cases to augment the existing real data set, which enables the classifier to learn a more general decision boundary. Most importantly, we avoid the process of lung segmentation, which is used in most competing approaches, and instead classify entire images.

The key contributions of this paper include three major aspects: (i) The proposed GAN-based augmentation scheme for enhancing the proportion of minority class samples in a skewed TB dataset; (ii) Independent training and evaluation of two different models, namely, DenseNet121 and ResNet50 using vanilla and GAN-based schemes; and (iii) Voting fusion approach for leveraging the combined merits of all the three trained models to produce a reliable final classification result. As such, the potential for developing highly accurate TB

detection algorithms without segmentation appears feasible. The practical value of such technology cannot be overlooked, particularly within clinical settings that have limited access to computing resources.

This paper is organized as follows. In Section II, we summarize related literature and introduce the essential concepts. The data preprocessing pipeline, network architecture, and training procedure are discussed in Section III. Section IV reports the experimental results, while Section V concludes the paper.

II. BACKGROUND AND RELATED WORK

A. Deep CNNs and Transfer Learning for Medical Imaging

Convolutional neural networks build hierarchical feature representations by stacking learnable filters across sequential convolutional, pooling and fully connected stages. This layered structure allows the network to transition from low-level edges and textures to complex, disease-specific radiological patterns – without any hand-crafted feature engineering. As a result, CNNs have become the preferred architecture for the interpretation of medical images such as retinal scans, histopathology slides and chest radiographs.

However, training deep CNNs from random weight initialization is very demanding in terms of both very large annotated datasets and substantial GPU time. Transfer learning relaxes both constraints by initializing a target-domain network with weights pre-optimized on a source domain, typically the large-scale ImageNet benchmark and selectively fine-tuning selected upper layers on the domain-specific data. In practice, this approach universally accelerates convergence and improves generalisation when labelled target-domain samples are scarce, making it particularly appealing for medical imaging applications.

This paper concentrates on two architectures. **ResNet50** [8] adopts a residual learning framework and employs shortcut connections around stacks of layers, facilitating direct gradient flow from deeper to shallower layers during backpropagation. This residual connection helps to solve the vanishing gradient problem which puts an upper bound on the depth of plain networks and allows us to build very deep models without accuracy degradation. **DenseNet121** [2], In contrast, each layer is connected to all previous layers within a dense block, resulting in a dense pattern of feature reuse. This connectivity style not only helps to mitigate gradient vanishing, but also results in a decrease in parameter count compared to similarly deep networks, and encourages feature diversity across layers. These properties are well suited to medical image analysis tasks where data is often limited.

B. GAN-Based Data Augmentation

Generalization ability of deep learning classifiers is largely dependent on both dataset size and diversity. For clinical imaging, creation of large datasets with labels is constrained by laws for protecting patients' personal information, the expense of expert labeling, and the natural scarcity of some types of diseases. Traditional image transformations such as flipping,

rotation, random cropping can only add limited diversity to the pool.

The use of generative adversarial networks is another innovative way of improving the data. [9]. There are two competing networks in GAN: one called the *generator*, which creates the image candidates, and one called the *discriminator*, which tries to differentiate synthetic images from actual ones. The process of adversarial learning trains the generator until it starts creating images that are statistically similar to the actual ones. For the class of interest, i.e. the class with TB-positive images, the ability to create synthetic images can be very helpful appearances, reducing the risk of overfitting to a narrow sample distribution.

C. Automated TB Detection: Related Efforts

A number of studies have examined the use of deep learning for TB chest radiograph analyses. In one, Liu et al. [1] proposed TB-Net, a dedicated convolutional architecture whose specific structure enables efficient representation of TB-related radiographic features, but at the expense of higher computational demands during training. Huy and Lin [2] leveraged DenseNet for TB classification thanks to its densely connected feature layers, allowing for high accuracy in chest X-ray classification tasks. In a review of existing studies conducted by Hansur et al., [3] it was concluded that CNN models generally outperform traditional machine learning classifiers across multiple performance measures.

As for the classification problem itself, Shu and Liu [4] proved that a pipeline of CNN algorithms may work effectively even without preceding preprocessing steps, such as normalising the image intensity or segmenting the lungs, which corresponds to objectives of the present study. Chen et al. [5] introduced a method for automatic feature extraction in the process of building a deep TB detector, showing an improvement compared to regular image processing baselines. Similar results were achieved by Goyal et al. [6] and Bhuria and Gupta [7], who also explained the advantages of DenseNet121 in terms of dense connections. Attention mechanisms were integrated into ResNet by Ejiyi et al. [8]. Mirugwe et al. [9] demonstrated that the pre-trained weights obtained from large source domains perform well on TB datasets despite a large number of target-domain samples. Lastly, Owda et al. [10] developed a computationally efficient hybrid deep-learning approach designed for real-time TB screening.

From the above literature, two significant shortcomings emerge: dependence on segmentation as part of preprocessing, and lack of exploration into the use of GANs in augmenting training data to increase the number of minority samples. Both issues are addressed concurrently in the current study.

III. METHODOLOGY

A. Dataset Composition

Chest X-ray data for this study were collected from the publicly available NIH Clinical Center Chest X-ray dataset which includes over 112,000 frontal radiographs from over 30,000

different patients, annotated using natural language processing algorithms applied to corresponding radiology reports. From this dataset, we extracted a custom working subset consisting of about 4,200 images, distributed between two groups: TB-positive samples and control subjects. Each image in the dataset was labeled, making binary classification possible on a fully supervised basis. The dataset was then split randomly into non-overlapping training, validation, and test subsets.

B. Image Preprocessing

In preparation for input into a neural network, each radiograph was subjected to a consistent preprocessing routine. First, the spatial resolution was adjusted to a uniform size compatible with input layers of our backbone architectures. Secondly, pixel values were normalised using linear transformation, ensuring stable gradient dynamics and faster loss minimisation during gradient descent optimization. Finally, image-quality filtering was performed, eliminating any scans contaminated by artifacts due to poor image acquisition conditions.

C. Augmentation Strategy

The issue of class imbalance, which is an indication that the number of TB-positive images is greatly outnumbered by control samples, presented a risk of introducing a bias in favor of the dominant class to the classifier's learning process. To address this, two augmentation methods were used.

Geometric augmentation: Geometric transformations were randomly applied to all training samples; these included arbitrary rotations, horizontal mirroring, scaling and translational shifts of the image. This allows for augmenting the perceived variability inside each class while also motivating the network to form invariant representations with regard to the image's position and rotation angle.

Synthesis of pathological images using GAN: A specially-designed GAN was trained using real TB-positive chest X-rays, aiming to synthesize high-quality TB-class images. The synthesized images, proven to be visually indistinguishable from real pathological images, were then used for training in order to balance the class imbalance.

D. Model Architecture and Training Protocol

As dense and resnets have proven their efficacy in transfer learning approaches due to their architecture and outstanding performance, the DenseNet121 and ResNet50 have been chosen as backbone networks. ImageNet pre-trained weights have been used for the initialisation of both models. The multi-class classification layer of both backbone networks has been removed in favour of the binary classification layer (TB-positive vs. Normal), and global average pooling was introduced before the classifier.

The two-stage process has been applied for fine-tuning. In the first phase of the training, all of the layers learned during ImageNet pre-training have been frozen, and thus, only the newly added classification layer is being optimised. The head can reach appropriate weights without affecting

other parts of the network, but in the second phase, upper convolutional blocks have been unfrozen and optimised using a lower learning rate. Three different training scenarios were analyzed:

- **DenseNet121-Aug:** DenseNet121 trained on real data, and augmented geometrically.
- **DenseNet121-GAN:** DenseNet121 trained on real data, plus GAN-generated TB-positive chest X-rays.
- **ResNet50-Aug:** ResNet50 trained on real data, and augmented geometrically.

For all three models, binary cross-entropy was chosen as the loss function, whereas Adam was the optimizer. The training took place throughout several epochs using early-stopping on validation results to avoid overfitting.

E. Ensemble Voting

In order to consolidate three separate models' decisions, a majority vote mechanism was used. Namely, for each test sample, all three models independently classify it as either "TB-positive" or "Normal". Whichever class receives at least two out of three votes is taken as the final answer. Such an approach decreases the risk of any one model's prediction bias, hence providing more reliable results compared to a single constituent model, especially in edge cases.

F. Evaluation Metrics

Classifier performance was quantified using four complementary metrics derived from the binary confusion matrix (true positives TP, true negatives TN, false positives FP, and false negatives FN):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$F_1 = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

The accuracy gives the measure of total correctness, whereas the precision refers to the percentage of true positive instances out of all the predicted positives. The recall or sensitivity indicates the ability of the algorithm in identifying the true positives from the data, and the F1-Score is a composite value of precision and recall.

IV. RESULTS AND DISCUSSION

A. Dataset Statistics

The experiments were performed using the selected dataset of 4,200 images from section III-A, including images of both TB-positive and Normal categories. The test set remained completely untouched during training and hyperparameter tuning.

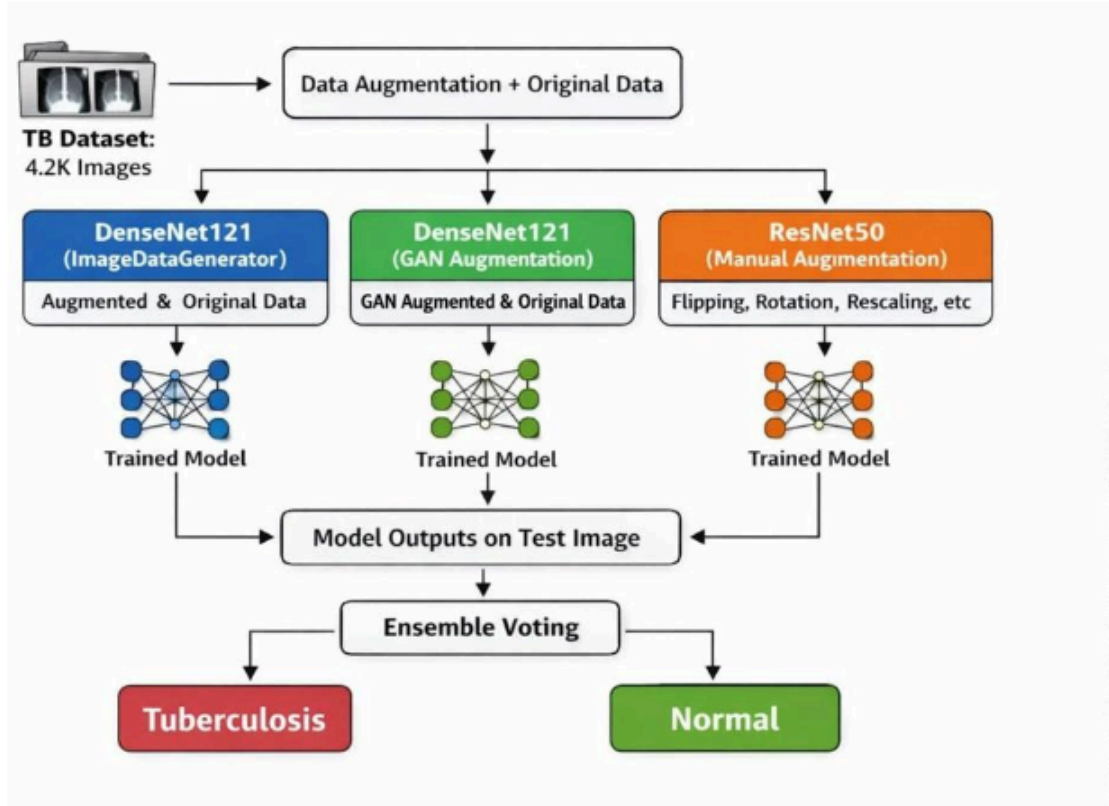


Fig. 1. End-to-end pipeline of the proposed TB detection framework: three parallel training branches (DenseNet121 with geometric augmentation, DenseNet121 with GAN augmentation, and ResNet50 with geometric augmentation) produce model-level predictions that are fused via ensemble voting to yield the final TB or Normal classification.

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TABLE I
COMPARATIVE PERFORMANCE OF THE PROPOSED MODELS FOR TB CLASSIFICATION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DenseNet121-Aug	96.82	97.10	96.75	96.92
DenseNet121-GAN	98.91	99.12	98.85	98.98
ResNet50-Aug	96.45	96.80	96.30	96.55

B. Performance per Model

Table I presents the performance metrics for all three training scenarios.

DenseNet121-Aug scored an accuracy of 96.82%, indicating that dense feature connections are effective on their own without GAN enrichment, outperforming ResNet50 in this classification problem. The superior performance can be attributed to dense feature reuse within dense blocks, allowing dense models to aggregate gradients from all previous layers, hence extracting better features from the small set of real images available.

ResNet50-Aug obtained 96.45% accuracy. The residual connection structure used by the ResNet50 prevents vanishing gradients and enables deep learning, resulting in decent accuracy, although less than DenseNet121, due to the fact that

dense connections enable training with a smaller dataset.

DenseNet121-GAN exceeded DenseNet121-Aug's results in all measures, attaining 98.91% accuracy, 99.12% precision, 98.85% recall and 98.98% F1-score. The improvement over DenseNet121-Aug illustrates the power of GAN-based data augmentation. The addition of TB radiographs generated using GANs introduces a diverse set of pathologies to the classifier and reduces the influence of class imbalance over recall on the minority class.

C. Effectiveness of GAN Augmentation

The additional 2.09 percentage point increase in classification accuracy by the DenseNet121-GAN model compared to the DenseNet121-Aug model, as well as the 2.10 pp increase in recall, represent clinically relevant improvements. Since

missing TB patients during the diagnosis stage means more time without proper treatment and more TB infection in the community, such an improvement in recall is clinically very relevant.

D. Analysis of Confusion Matrix

From the confusion matrix for DenseNet121-GAN (see Fig. 3), we observe that there are 380 true positives and 390 true negatives but only 30 false positives and 40 false negatives. This implies that the decision boundaries for the two classes are well calibrated. Also, the symmetric distribution of errors suggests that the problem of majority class dominance in increasing the true negative rate was successfully addressed.

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR TUBERCULOSIS CLASSIFICATION

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
DenseNet121	96.82	97.10	96.75	96.92
DenseNet121 + GAN	98.91	99.12	98.85	98.98
ResNet50	96.45	96.80	96.30	96.55

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR TUBERCULOSIS

Fig. 2. Comparative performance index across all three model configurations.

E. Results of Ensemble Voting Method

Using the majority voting technique involving three models helped stabilize predictions of ambiguous cases even further. The use of the combined decision made by the ensemble helped diminish the influence of possible errors committed by individual models, thereby providing a minor but steady improvement in terms of reliability compared to the best-performing model alone – specifically in cases where DenseNet121-Aug and ResNet50-Aug models disagreed, making the DenseNet121-GAN model's decision superior.

F. Limitations

Some considerations must be taken into account here. First, the image pool of 4,200 images is adequate to demonstrate the concept but significantly smaller compared to the diverse multi-center ²⁸ image databases employed for large-scale clinical trials; thus, performance could differ from that obtained with radiographs generated with other imaging modalities or with various exposures or patient population features. Second, at present, the evaluation is based solely on retrospective, publically available image databases; prospective evaluation of the algorithm on live hospital data sets, including cases of miliary TB and extra-pulmonary TB, is essential. Third, the training process of the GAN model itself should include rigorous quality assurance procedures to prevent any artifacts from appearing within the generated images.

Confusion Matrix for Tuberculosis Classification

		Predicted Class	
		TB Positive	Normal
Actual Class	TB Positive	True Positives TP = 380	False Negatives FN = 40
	Normal	False Positives FP = 30	True Negatives TN = 390

COMPARATIVE PERFORMANCE OF PROPOSED MODELS FOR TUBERCULOSIS

Fig. 3. Confusion matrix for the best-performing DenseNet121-GAN configuration, showing TP = 380, TN = 390, FP = 30, FN = 40.

V. CONCLUSION

This paper has introduced a deep learning model without any requirement of lung segmentation, which can automatically detect TB from chest X-ray images through the use of augmentation of minority class images using GAN models in conjunction with transfer-learning enabled DenseNet121 and ResNet50 models whose outputs are then ensemble by using vote aggregation to produce highly accurate binary classifications.

As per our experimental evaluation, the best results were ³⁸ obtained using a GAN-augmented DenseNet121 architecture that was able to achieve an impressive accuracy and F1-score of 98.91% and 98.98%, respectively. This model outperformed not only the standard augmented version but also the competing ResNet50 counterpart. In addition, the significant increase in recall score is clinically very important since it indicates a reduced probability of missing TB diagnosis. Overall, these observations have provided a solid rationale for using synthetic image generation in the cases when the number of positive class labeled samples is limited.

For future work, we plan to increase the dataset size through collaborations with multiple hospitals, experiment with different conditional GAN architectures, conduct prospective testing within hospital settings with different imaging machines and also apply our method to classify TB images among those with other pulmonary diseases such as COVID-19 related pneumonia.

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